

A 0D pulsatile model of aortic-arch banding

Reproducing carotid pulsatility redistribution (Eberth et al., 2009)

svZeroDSolver

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The problem

Eberth et al. (2009): a stiff band on the mouse aortic arch, placed *between* the two carotid takeoffs, creates paired carotids with **similar mean pressure & flow but very different pulsatility**.

- **RCCA-B** (right) — *upstream* of the band: sees the full pulse + reflections $\Rightarrow$ pulsatility index rises ($PI \approx 3.11$).
- **LCCA-B** (left) — *downstream*: the resistive–inertial band damps the pulse ($PI \approx 1.65$); baseline CCA ≈ 1.16 .

Goal: a reduced-order (0D) lumped model that reproduces this redistribution with the band as the only structural change — and that maps to the wall growth & remodeling Eberth measured.

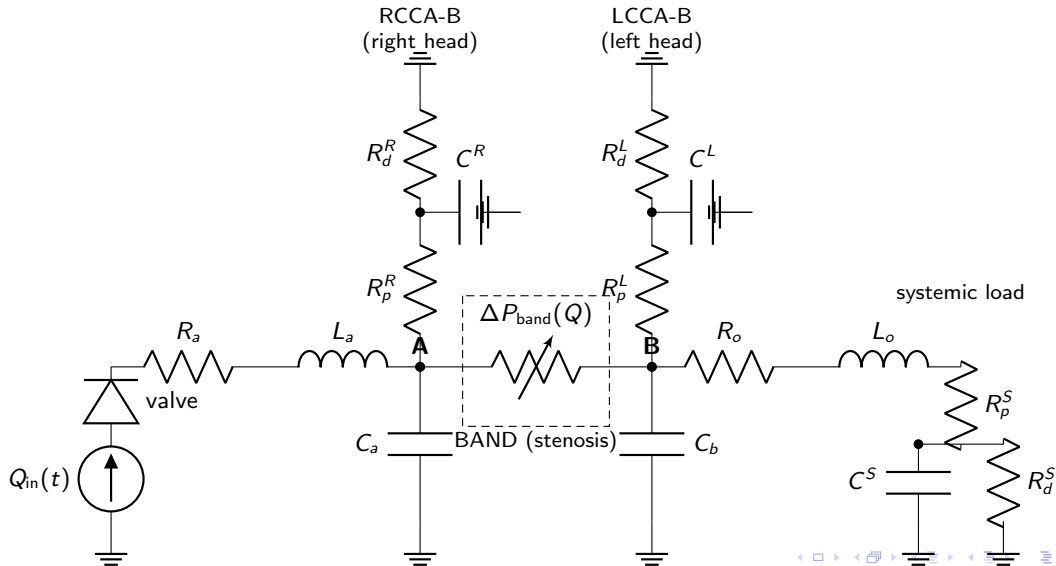
The 0D model

Heart $\rightarrow$ aortic valve $\rightarrow$ ascending aorta $\rightarrow$ **node A** $\rightarrow$ band $\Delta P(Q) \rightarrow$ **node B** $\rightarrow$ descending aorta;
a carotid RCR bed hangs off each node.

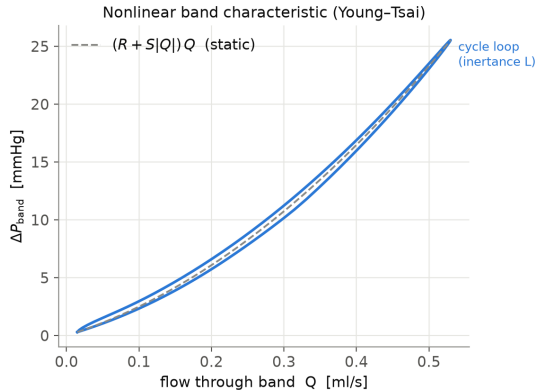
- Vessels: RCL BloodVessel blocks
- Band: nonlinear *Young-Tsai* stenosis
$$\Delta P = (R + S|Q|)Q + L \dot{Q}$$
- Terminal beds: 3-element RCR windkessels
- Valve: ValveTanh diode
- Parameters from mouse literature (geometry, blood, wall stiffness E)
- Every parameter tagged
FIXED / DERIVED / CALIBRATE
- Only the band `stenosis_coefficient` is calibrated (to the Table-1 PI ratio)

Three states differ *only* in carotid geometry: pre-banding (baseline), acute (baseline walls), chronic (Table-1 remodeled walls).

0D circuit (electric analogy)



The band is a nonlinear (Young–Tsai) element



ΔP – Q across the band: the quadratic Bernoulli/turbulent term ($S|Q|Q$) plus an inertance loop ($L \dot{Q}$). This nonlinearity + inertia is what damps the downstream pulse while leaving the upstream node pulsatile.

Parameter roles: fixed, assumed, and the knob we tune

Fixed (measured / literature)

- blood $\rho = 1.06$, $\mu = 3.5$ cP
- wall E : aorta 1.0, carotid 1.5 MPa
- HR 6.09 Hz, CO 0.20 ml/s, MAP 90
- carotid ID/wall (Table 1), aorta dia.
- measured central aortic compliance

⇒ all vessel R , L , C are *derived* from these via Eq. (1)–(3).

Assumed (plausible, not fitted)

- segment lengths (carotid, band) — estimates
- ejection waveform shape (systolic fraction)
- RCR split $R_p:R_d = 10:90$; time const. τ
- distal pressure $P_d = 0$
- valve R_{max}/R_{min} /steepness

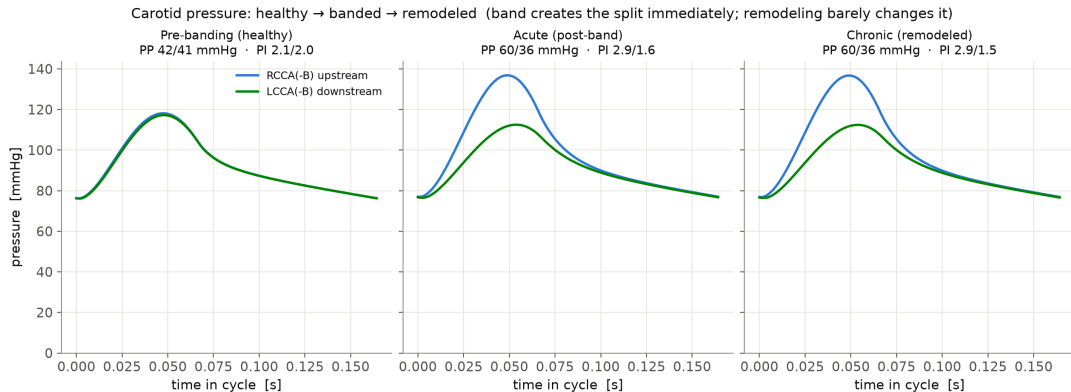
chosen by convention/estimate — could be refined, but not tuned to the data.

Knob (calibrated to data)

- band stenosis_coefficient S
- the **one** parameter we turn
- geometry gives a prior; tuned to the Table-1 PI ratio (3.11/1.65)

Everything else is fixed or derived — this is the only free knob fitted to the data.

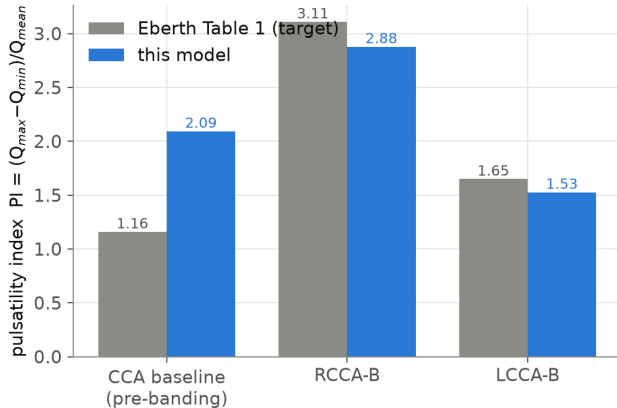
Result: banding creates the split immediately



Carotid pressure is **symmetric before banding**, splits **immediately** on banding (upstream $\uparrow$, downstream damped), and **barely changes** with wall remodeling $\Rightarrow$ the split is set by band *position*; remodeling is the response, not the cause.

Result: pulsatility vs. the paper

Pulsatility by location — calibrated model vs paper Table 1
(stenosis calibrated to the PI ratio; both PIs within ~7%)



	Paper	Model
RCCA-B PI	3.11	≈ 2.9
LCCA-B PI	1.65	≈ 1.5
PI ratio	1.89	1.89
PP RCCA-B	56	60
MAP (R/L)	similar	98/91

The **banded** carotids, the PI ratio, MAP similarity, and pulse pressure are reproduced; mean carotid flows land in the measured range.

Limitation: baseline vs. banded cannot both be matched in 0D

- The model **over-predicts the baseline (pre-banding) PI** (≈ 2.1 vs. the paper's 1.16), though baseline *pulse pressure* matches (≈ 41 vs. 42).
 - A parameter sweep shows every knob that lowers the baseline PI to 1.16 **also collapses the banded PI** — the passive bed is a linear pulsatility filter.
 - The paper's baseline→banded amplification is $\sim 2.7\times$ ($1.16\rightarrow 3.11$); a lumped 0D band delivers only $\sim 1.5\times$.
 - The missing $\sim 2\times$ is **wave reflection** off the band — a spatial/timing effect a 0D element cannot represent.
- ⇒ Reproducing the *full* amplitude needs a **1D** (wave-propagation) model — exactly the 0D→1D boundary.

Knobs & targets, by group

Control group (CCA)

- HR 7.17 Hz; baseline carotids; no band (*fixed*)
- bed R pinned to $\bar{Q}$ 0.016, MAP 92
- aortic $C = 2.67\text{e-}4$ (measured)
- **bed** $C = 6.3\text{e-}6 \rightarrow \text{PI } 1.16$

DOF: 4 targets / 4 knobs

Shared invariants (both groups): blood μ, ρ , topology, ejection shape, wall moduli E . **Bold** = calibrated knob $\rightarrow$ its target; everything else is fixed or pinned by measured data.

Banded group (RCCA-B / LCCA-B)

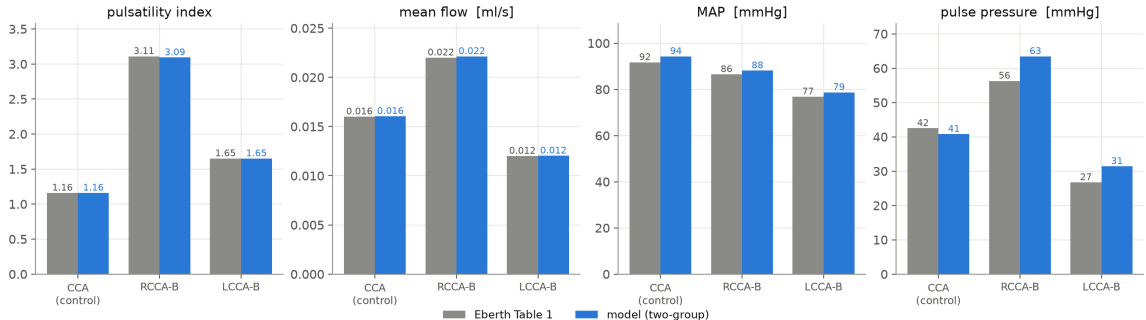
- HR 6.09 Hz; remodeled carotids; band (*fixed*)
- bed R pinned to $\bar{Q}$ 0.022/0.012, MAP 86/77
- **band** $S = 95 \rightarrow A \rightarrow B$ drop ≈ 10 mmHg
- **RCCA-B bed** $C = 2.5\text{e-}5 \rightarrow \text{PI } 3.11$
- **LCCA-B bed** $C = 2.0\text{e-}5 \rightarrow \text{PI } 1.65$

DOF: 8 targets / 8 knobs

Revised: two-group calibration (control vs. banded)

Control (CCA) and banded (RCCA-B/LCCA-B) are **two separate animal groups** — they need not share $R/C/L$. Calibrating each on its own terms — HR and measured flows/pressures pin the beds, the band anchored to the measured $A \rightarrow B$ pressure drop, per-carotid bed compliance sets the PI — matches *all three* carotid states with a stock RCR:

Two-group calibration vs Eberth Table 1 (control CCA + banded RCCA-B/LCCA-B)



DOF-balanced (8 targets / 8 knobs). Cost: bed compliance differs between groups (fitted, not measured) — descriptive, not a single predictive model.

Summary

- A literature-parameterized 0D model reproduces the Eberth **pulsatility redistribution**: upstream carotid pulsatile, downstream damped, at matched mean pressure/flow.
- Banding creates the split **immediately**; wall remodeling barely changes it — consistent with pulsatility being the *stimulus* for remodeling.
- With a *single shared* parameter set, 0D matches the redistribution and pressures but not both the baseline and banded PIs (a bounded amplification limit; a 1D wave-reflection effect is small for the mouse).
- Treating control and banded as *separate groups* — their own beds, band anchored to the measured pressure drop — matches **all three carotid states**, at the cost of group-specific (fitted) bed compliance.